Saakshi Wadhwa

Nationality - India | Residence - 20 Bounty Street, Springfield lakes, QLD 4300
+61469798287 saakshi.wadhwa@unisq.edu.au Saakshi Wadhwa

EDUCATION

University of Southern Queensland **April 2026 - Present**
PhD in Asteroseismology *Springfield, Queensland, Australia*

KU Leuven **2023 - 2025**
Masters of Astronomy and Astrophysics- ***Score - 72.1 % (Cum Laude)*** *Leuven, Belgium*
Relevant Coursework - Star Formation, Observational Techniques in Astronomy, Binary Stars, Stellar Structure and Evolution, Radiation Processes in Astronomy.

Fergusson College **2020 – 2023**
Bachelor's of Science in Physics - ***CGPA - 8.93/10*** *Pune, India*

ACHIEVEMENTS

- Secured a **Silver Honour** in IAAC Exam 2021.
- Secured Second Prize in Poster exhibition competition for two consecutive years, arranged by Astro Club, Fergusson College.
- Ranked in the top 1 % in Maharashtra 12th Grade Exams.
- Made **1 Provisional and 8 Preliminary discoveries of Asteroids** in an International Asteroid Search Campaign in August 2021.

RELEVANT PROJECTS

Pre-main sequence pulsators in the Perseus Molecular Cloud **KU Leuven**
Master's Thesis *- Dr. Dario Fritzewski* *2024-25*

- Conducted asteroseismic analysis of **young pre-main sequence pulsating stars** in the Perseus molecular cloud using TESS full-frame image data.
- Identified **8 Delta Scuti stars and 1 Gamma Dor** star from a sample of 150 stars, along with 3 candidate g-mode pulsators.
- Successfully identified the **second ever** pre-main sequence Gamma Dor star with a distinct period spacing pattern.

Construction of a Spectral Energy Distribution **KU Leuven**
Course - Observational Techniques in Astronomy *2023*

- Constructed and studied the SED of a **post-AGB star with circumstellar dust** located in the large Magellanic Cloud.
- I further tried to extract physical properties of the star and the dust around it using the SED.

Photometric Follow-up Observations of Variable Stars **KU Leuven**
Course - Research Projects - Dr. Gang Li *2023*

- Re-classification of a set of 307 variable stars that were previously observed by the ground-based telescope, CSTAR.
- Analysed the light curves and the frequency spectra of each star using photometric data from the space-based TESS telescope.

Stellar Evolution and Analysis **KU Leuven**
Course - Binary stars, Stellar Structure and Evolution *2023-24*

- Performed **single and binary star evolution simulations using the MESA Software** and tried to determine the physics behind their evolution.
- Performed **Binary Star Analysis** and Evolution with PHOEBE and MESA.

PROJECTS

Interferometry of Massive Stars **KU Leuven**
Course - Research Projects in theoretical Astrophysics *2024*

- Analysis of interferometry data obtained from the PIONIER instrument at VLTI, successfully revealing companions of two unpublished O type stars.

Evolution of Galaxy Structure over Cosmic time **KU Leuven**
Course - Galaxies and Cosmology *2024*

- Reading Project along with original research on the topic.
- **Performed double Sersic fits** for different galaxy models using Statmorph.

Forecasting of CMEs and the CME-driven shock wave arrival to Earth **KU Leuven**
Course - Space Weather *2024*

- Analyzed solar radio observations of a Type II burst using polarization data, density models, and EUHFORIA simulations to study CME dynamics and propagation, comparing results with in-situ OMNI datasets.

Developing an Automated Imaging Pipeline for GMRT **Fergusson College**
Bachelor's Thesis *- Dr. Mayuresh Surnis* *2022-23*

- Modified the CAPTURE pipeline in the Common Astronomy Software Applications (CASA) package to automate radio imaging using data from the Giant Metrewave Radio Telescope (GMRT), ensuring compatibility with both the legacy GMRT and the upgraded uGMRT.

COMMUNITY SERVICE & OUTREACH

- Volunteered to conduct a spectroscopy workshop for high-school student visits at KU Leuven.
- Volunteered and Organized 3 editions of Frontiers in physics(2021-23), A 2-day national level event hosted by Astro Club, Fergusson College.
- Volunteered for National Science Day 2023 in Inter University Centre for Astronomy and Astrophysics wherein I explained the formation of stars to more than a 1000 people.
- Conducted a star-gazing workshop for classes 3-5, at DLRC, Pune in November 2022.
- Conducted 1-day workshops on Planetary systems in various schools around Pune on the occasion of National Science Day 2022.

EXTRACURRICULAR

- Student Representative for the Masters of Astronomy and Astrophysics program at KU Leuven.
- Volunteered and Organised numerous national level events and workshops related to the field of Science at the Astro Club of Fergusson College.
- Member of Habitat for Humanity organization since my school days. I have volunteered in many activities like house builds, charities and marathons.
- Worked as a member of Saathi's Executive Team (an NGO for blind students in Fergusson College). We recorded academic and competitive exam reference books for students, provided writers, supported English improvement, and assisted in their studies.

COMPUTATION KNOWLEDGE & SKILLS

- **Python**
- **MESA**
- Stellarium
- Anchoring
- LaTeX
- MongoDB Basics
- CASA
- Management
- LITPro
- FEKO
- **PHOEBE**

LANGUAGES

- English, Hindi